Rocket Nozzle Simulation

OpenFOAM v2012 · rhoCentralFoam · Axisymmetric wedge · Compressible supersonic flow

A parametric study of flow through a De Laval (converging-diverging) rocket nozzle, comparing an idealised hot-gas model against a reacting LOX/kerosene combustion case. Both cases use the same axisymmetric wedge geometry and are run with density-based compressible solvers capable of capturing the transonic throat, supersonic expansion, and shock structure at the nozzle exit.

Cases

Case	Solver	Inlet model	Status
Hot gas (frozen composition)	`rhoCentralFoam`	Combustion products approximated as a perfect gas (M=22 g/mol, gamma=1.2)	Complete
LOX/kerosene reacting flow	TBD	Full combustion chemistry with species transport	In progress

Nozzle Geometry

All cases share the same axisymmetric conical De Laval profile:

Parameter	Value
Inlet radius	0.150 m
Throat radius	0.050 m
Exit radius	0.1414 m
Throat position	x = 0.15 m
Total nozzle length	0.50 m
Converging area ratio	9:1
Diverging area ratio	8:1
Design exit Mach (gamma=1.2)	~3.2

The mesh is a 5-degree wedge (axisymmetric) built with `blockMesh`, 40 radial cells x 150 axial cells = 6,000 cells total. The two-block structure (converging + diverging) uses `wedge` boundary patches on the front and back faces and an `empty` patch on the degenerate axis line.

Hot Gas Case

Setup

Combustion products of LOX/kerosene are approximated as a single-species perfect gas with properties representative of the chamber exit composition:

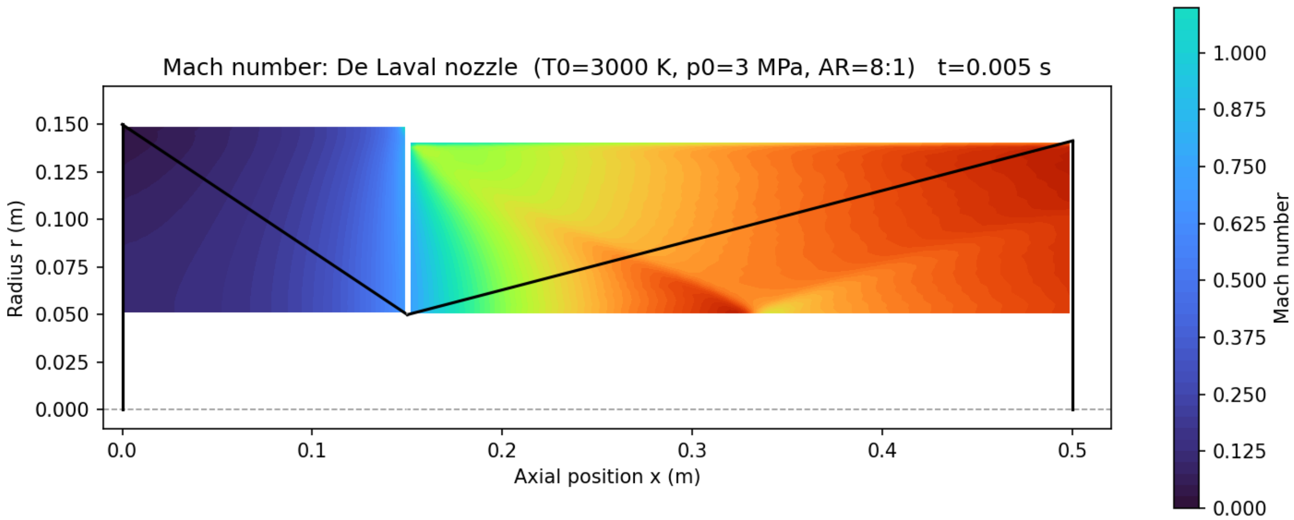
Property	Value	Notes
Molecular weight	22 g/mol	Mix of CO_2, H_2O, CO, H_2
gamma	1.2	Tri-atomic dominated at high T
Cp	2267 J/(kg K)	Derived from gamma, M
Dynamic viscosity	1e-4 Pa s	Approximat at 3000 K
Chamber total pressure	3 MPa	
Chamber total temperature	3000 K	Approximat LOX/kerose adiabatic flame temperature
Ambient back pressure	101,325 Pa	Sea level — nozzle is over-expan at design

Boundary conditions: `totalPressure` and `totalTemperature` at the inlet, `waveTransmissive` at the outlet. The case is run laminar (viscous effects are secondary for this geometry and Reynolds number).

Solver: `rhoCentralFoam` with Kurganov-Tadmor flux scheme, MUSCL reconstruction (`vanLeer` for rho and T, `vanLeerV` for U), CFL = 0.1. Initialised with an isentropic pressure and temperature ramp along x to avoid the numerical temperature inversion that occurs when starting from uniform chamber conditions against a near-vacuum outlet.

Results

Mach number contour



The flow accelerates from rest at the inlet, reaches sonic conditions (Mach 1) at the throat (x = 0.15 m), then expands supersonically through the diverging section to a peak exit Mach of **3.22**, matching the isentropic design prediction of 3.2 to within 1%.

Quantity	Simulated	Isentropic theory
Exit Mach	3.22	3.20
Exit static T	1488 K	1482 K
Peak exit velocity	2667 m/s	2637 m/s

The oblique feature visible near x = 0.2 m is an expansion fan originating from the sharp throat corner — a real compressible flow effect, not a numerical artefact. The nozzle is over-expanded at sea level (design exit pressure ~43.5 kPa vs 101.3 kPa ambient), which would produce oblique shocks outside the nozzle exit in a full domain; those are outside the current computational domain.

LOX/Kerosene Case

(In progress) — Full reacting flow with species transport and combustion chemistry, replacing the frozen-composition inlet condition with actual LOX/kerosene combustion. Same nozzle geometry.

Workflow

```
blockMesh (5-degree wedge, 6000 cells) | rhoCentralFoam (density-based, explicit, CFL=0.1) |
Post-processing: Mach = |U| / sqrt(gamma * R_spec * T)
```

Key solver notes:

- `rhoCentralFoam` requires `fluxScheme Kurganov` and `vanLeer` interpolation schemes — these are not inherited from `fvSchemes` defaults and must be set explicitly.
- `totalTemperature` BC in v2012 requires a `gamma` entry in the field file, not just in `thermophysicalProperties`.
- Initialising with uniform chamber conditions causes temperature inversion in the first timestep due to the large pressure gradient across the domain. An isentropic ramp initialisation resolves this.
- The wedge block vertex ordering must ensure direction 3 (the extrusion direction, v0 to v4) is a non-degenerate edge — starting both groups with a wall vertex rather than an axis vertex avoids the "inward-pointing faces" error from the degenerate zero-length axis edge.

Limitations

- Conical nozzle profile — a bell or Rao-optimised profile would reduce exit flow divergence angle and improve specific impulse.
- Adiabatic wall assumption — no heat transfer to the nozzle wall.
- Laminar — boundary layer transition and turbulent mixing near the wall are not modelled.
- 2D axisymmetric — no three-dimensional instabilities or asymmetric shock structures captured.
- The over-expanded exit shock structure forms outside the domain.

Software Stack

Tool	Version	Purpose
OpenFOAM	v2012	CFD solver
ParaView	5.11	Post-processing
Python / matplotlib	3.x	Mach contour plot
Ubuntu (WSL2)	24.04	OS

Repository

Part of the [OpenFOAM Portfolio], a collection of CFD simulations demonstrating thermal and fluid simulation skills relevant to thermal engineering roles.